

Inference for Variance (UPDATED)

CSU, Chico Math 351

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outline

Testing Variance
Confidence Intervals
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Why σ^2 ?

When random samples are drawn from a normal distribution, the parameter μ is typically the target of the investigation. We would thus construct a confidence interval for μ or perform a hypothesis test involving μ .

However situations do arise when we are interested in the “precision” of the data. If this is the case we need to shift our focus on the scale parameter, σ^2 .

Properties of σ^2

1. The sample variance S^2 is an unbiased estimator for σ^2 and it is based on the maximum likelihood estimate.

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2$$

2. The ratio:

$$\frac{(n-1) S^2}{\sigma^2} = \frac{1}{\sigma^2} \sum_{i=1}^n (Y_i - \bar{Y})^2$$

has a Chi square distribution with $n-1$ degrees of freedom. We will use these two facts to make inferences about the parameter σ^2 .

With these properties, we can make inferences about the parameter σ^2 .

Constructing Confidence Intervals

Since the ratio $\frac{(n-1)S^2}{\sigma^2}$ has a chi square distribution with $n - 1$ degrees of freedom, we can write:

$$P \left[\chi_{\alpha/2, n-1}^2 \leq \frac{(n-1)S^2}{\sigma^2} \leq \chi_{1-\alpha/2, n-1}^2 \right]$$

We then just need to isolate the σ^2 just like we did when we were constructing a confidence interval for μ .

Constructing Confidence Intervals

Let s^2 denote the sample variance calculated from a random sample of n observations drawn from a normal distribution with mean μ and variance σ^2 . Then

1. A $100(1 - \alpha)\%$ confidence interval for σ^2 is the set of values

$$\left[\frac{(n-1)s^2}{\chi_{1-\alpha/2, n-1}^2}, \frac{(n-1)s^2}{\chi_{\alpha/2, n-1}^2} \right]$$

2. A $100(1 - \alpha)\%$ confidence interval for σ is the set of values

$$\left[\sqrt{\frac{(n-1)s^2}{\chi_{1-\alpha/2, n-1}^2}}, \sqrt{\frac{(n-1)s^2}{\chi_{\alpha/2, n-1}^2}} \right]$$

One-Sided Confidence Intervals

One-sided cases will work similarly to those for the t-distribution, but be mindful of the denominator since these confidence intervals are for σ^2 , not for the χ^2 distribution.

Let s^2 denote the sample variance calculated from a random sample of n observations drawn from a normal distribution with mean μ and variance σ^2 . Then

1. A $100(1 - \alpha)\%$ LOWER confidence interval for σ^2 is the set of values

$$\left[0, \frac{(n-1)s^2}{\chi_{\alpha, n-1}^2} \right]$$

2. A $100(1 - \alpha)\%$ UPPER confidence interval for σ^2 is the set of values

$$\left[\frac{(n-1)s^2}{\chi_{1-\alpha, n-1}^2}, \infty \right]$$

RStudio & Chi-Square values

Just as

$$z_{1-\alpha}^* = \text{qnorm}(1-\alpha)$$

and

$$t_{1-\alpha, n-1}^* = \text{qt}(1-\alpha, n-1)$$

we can use built-in R functions in place of distribution tables to find the values we need. Be sure to match the subscript and your arguments in the function:

$$\chi_{1-\alpha, n-1}^2 = \text{qchisq}(1-\alpha, n-1)$$

Test Statistic

Let s^2 denote the sample variance calculated from a random sample of n observations drawn from a normal distribution with mean μ and variance σ^2 .

For a hypothesis test with $H_0 : \sigma^2 = \sigma_0^2$, the test statistic is then:

$$\chi^2 = (n - 1) s^2 / \sigma_0^2.$$

We reject the null hypothesis if the p -value is less than the confidence level α .

p-values

We can use the `pchisq()` function to calculate p-values for the χ^2 Distribution, just as we used `pnorm()` and `pt()` for their respective distributions.

However, since this is **NOT** a symmetric distribution, $\alpha/2$ and $1 - \alpha/2$ percentiles will not be opposites. Thus, calculation will be a bit different for two-sided tests (and won't use `abs()`).

1. To test $H_0 : \sigma^2 = \sigma_0^2$ vs. $H_1 : \sigma^2 < \sigma_0^2$, use:

$$\text{p-value} = \text{pchisq}(\chi^2, n-1)$$

2. To test $H_0 : \sigma^2 = \sigma_0^2$ vs. $H_1 : \sigma^2 > \sigma_0^2$, use:

$$\text{p-value} = 1 - \text{pchisq}(\chi^2, n-1)$$

3. To test $H_0 : \sigma^2 = \sigma_0^2$ vs. $H_1 : \sigma^2 \neq \sigma_0^2$, use:

$$\text{p-value} = 2 * \min(\text{pchisq}(\chi^2, n-1), 1 - \text{pchisq}(\chi^2, n-1))$$

Alternative Approach: When to Reject

Alternatively, if we just care about whether we reject H_0 or not, we can use the critical values from earlier to define our cutoffs:

1. To test $H_0 : \sigma^2 = \sigma_0^2$ vs. $H_1 : \sigma^2 < \sigma_0^2$:

Reject H_0 if $\chi^2 \leq \chi_{\alpha, n-1}^2$.

2. To test $H_0 : \sigma^2 = \sigma_0^2$ vs. $H_1 : \sigma^2 > \sigma_0^2$:

Reject H_0 if $\chi^2 \geq \chi_{1-\alpha, n-1}^2$.

3. To test $H_0 : \sigma^2 = \sigma_0^2$ vs. $H_1 : \sigma^2 \neq \sigma_0^2$:

Reject H_0 if $\chi^2 \leq \chi_{\alpha/2, n-1}^2$ OR $\chi^2 \geq \chi_{1-\alpha/2, n-1}^2$.

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Example 1

The data set "mineral.csv" <https://raw.githubusercontent.com/njlytal/MATH351/refs/heads/main/mineral.csv>. contains potassium-argon estimated ages (in millions of years) for 19 mineral samples. Assume the estimated ages are normally distributed with unknown mean μ and variance σ^2 . Find a 95% confidence interval for the population standard deviation, σ .

Example 2

How much interest certificates of deposit (CDs) pay varies by financial institution and also by length of the investment. A large sample of national one-year CD and a standard deviation $\sigma=0.262$.

A five year CD ties up an investor's money, so it usually pays a higher rate of interest. However, higher rates might cause more variability. The table on the next slide lists the five-year CD rate offerings from $n=10$ banks in the northeast United States.

Find a 95% confidence interval for the standard deviation. Also, perform a hypothesis test to see if the five-year CDs are more variable than those for one-year certificates.

Example 2, Table

Bank	Interest Rate(%)
Domestic Bank	2.21
Stonebridge Bank	2.47
Waterfield Bank	2.81
NOVA Bank	2.81
American Bank	2.96
Metropolitan National Bank	3.00
AIG Bank	3.35
iGobanking.com	3.44
Discover Bank	3.44
Interwest National Bank	3.49

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Adapted from notes by Dr. Gray.